NumPy & Pandas Lab Programs

1. Student Average

Aim

To calculate the average marks of each subject and identify the subject with the highest average.

Algorithm

1. Start

2. Read Students_data.csv

3. Extract subject marks

4. Compute average of each subject

5. Find highest average subject

6. Display results

7. Stop

Code

```
import numpy as np

student_scores = np.loadtxt(
    "Students_data.csv",
    delimiter=",",
    skiprows=1,
    usecols=(2,3,4,5)
)

subjects=["Math","Science","English","History"]
average_scores=np.mean(student_scores,axis=0)
highest_subject=subjects[np.argmax(average_scores)]

print("Average scores:",np.round(average_scores,2))
print("Subject with highest average score:”,highest_subject)
```

Output

Result

Thus, the average marks of each subject were calculated successfully and the highest average subject was identified.

2. Past Month Price Average

Aim

To calculate the average price for the past month using Pandas.

Algorithm

1. Start

2. Read Sales_data.csv

3. Filter November 2023 records

4. Calculate average price

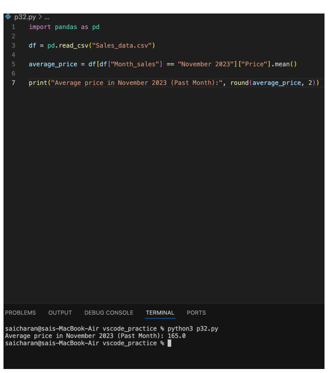
5. Display result

6. Stop

Code

```
import pandas as pd

df=pd.read_csv("Sales_data.csv")
average_price=df[df["Month_sales"]=="November 2023"]["Price"].mean()
print("Average price:",round(average_price,2))
```

Output



Result

Thus, the average price for the selected month was calculated successfully.

3. House Average Price

Aim

To calculate the average price of houses having more than 4 bedrooms.

Algorithm

1. Start

2. Read House_data.csv

3. Select houses with bedrooms > 4

4. Calculate average price

5. Display result

6. Stop

Code

```
import numpy as np

house_data=np.loadtxt("House_data.csv",
delimiter=",",
skiprows=1,
usecols=(1,2,3,5))

houses=house_data[house_data[:,0]>4]
average_price=np.mean(houses[:,3])
print("Average price:",round(average_price,2))
```

Output

Result

Thus, the average price of houses with more than four bedrooms was calculated successfully.

4. Quarterly Sales Analysis

Aim

To calculate the total yearly sales and percentage increase from Q1 to Q4.

Algorithm

1. Start

2. Read Sales_data.csv

3. Extract month and sales columns

4. Calculate Q1 and Q4 sales

5. Calculate yearly sales

6. Calculate percentage increase

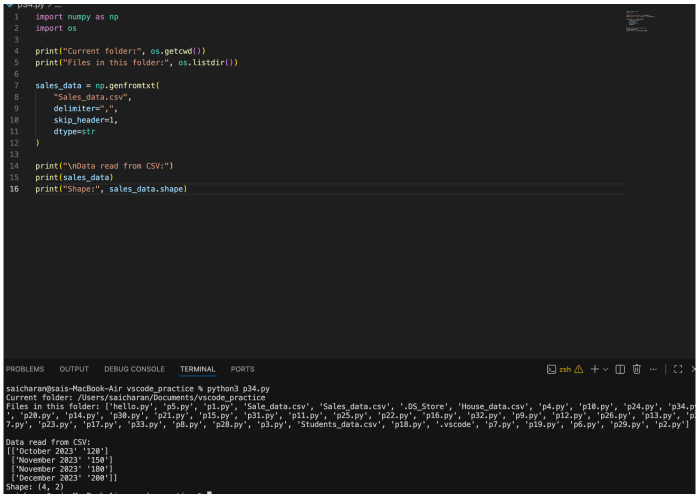
7. Display results

8. Stop

Code

```
import numpy as np

sales_data=np.genfromtxt("Sales_data.csv",
delimiter=",",
skip_header=1,
dtype=str)

months=sales_data[:,1]
sales=sales_data[:,4].astype(float)

Q1=0
Q4=0

for i in range(len(months)):
    if months[i] in ["January","February","March"]:
        Q1+=sales[i]
    elif months[i] in ["October","November","December"]:
        Q4+=sales[i]

total=np.sum(sales)
increase=((Q4-Q1)/Q1)*100

print("Total sales:",total)
print("Percentage increase:",round(increase,2),"%")
```

Output



Result

Thus, the yearly sales and percentage increase from Q1 to Q4 were calculated successfully.